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| <b>14</b> | A solid X is in thermal equilibrium with a solid Y, which is at the same temperature as a third solid Z. The three bodies are of different materials and masses. Which one of the following statements is certainly correct? |                                                                              |
|           | <b>A</b>                                                                                                                                                                                                                     | X and Y have the same heat capacity.                                         |
|           | <b>B</b>                                                                                                                                                                                                                     | Y and Z have the same internal energy.                                       |
|           | <b>C</b>                                                                                                                                                                                                                     | The atoms of X and Y have the same speed.                                    |
|           | <b>D</b>                                                                                                                                                                                                                     | There is no net transfer of energy if Y is placed in thermal contact with Z. |

**Ans: D**

All the bodies have the same temperature and hence have the mean kinetic energy. This does not mean they have the same internal energy which also depends on the mass of the body or number of atoms in the body. (A and B are not definitely true)

The atoms have the same root mean square speed, not the same speed so C is also not true.

D is correct as all bodies are in thermal equilibrium.

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| <b>15</b> | Equal masses of two solids X and Y are heated from the same temperature. Energy is |
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